MAHATMA GANDHI UNIVERSITY, KOTTAYAM

THIRD SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION – OCTOBER 2025

Third Semester

Discipline Specific Elective (DSE)-MG3DSECFS201-OPERATING SYSTEMS (Specialization: Operating System Architecture)

(2024 Admission Onwards)

Duration:2 Hours **Maximum Marks: 70**

SCHEME

Part A

Short Answer Type Questions

Answer any 10 questions.

Each question carries 2 marks.

1. Explain Batch Operating System with an example.
A batch operating system executes a batch of jobs together without manual intervention. Example: In early IBM mainframes, multiple jobs were grouped and executed sequentially.

2. Examine Mutual Exclusion.
Mutual exclusion ensures that only one process can access a shared resource at a time, preventing data inconsistency and race conditions.

3. Analyse Throughput in CPU Scheduling Efficiency.
Throughput measures the number of processes completed per unit time. Higher throughput means better CPU utilization and performance.

4. Define Segmentation in Memory Management.
Segmentation divides memory into logical units like code, data, and stack segments to allow flexible allocation and protection.

5. List any two Page Replacement Algorithms.
Examples: FIFO (First In First Out) and LRU (Least Recently Used).

6. Illustrate how Paging within each Segment Reduces External Fragmentation.
Paging divides segments into equal-size pages, fitting them into available frames—thus eliminating gaps in memory (external fragmentation).

7. Distinguish between Inode and Directory Entry.
Inode stores metadata about a file, while the directory entry links a filename to its inode number.

8. Analyse how Tree-Structured Directories reduce Name Conflicts.
Tree-structured directories organize files hierarchically, allowing different users to have identical filenames under different directories.

9. Examine the Relationship between Spooling and Concurrent Process Execution.
Spooling overlaps input/output operations with computation, allowing multiple processes to execute concurrently.

10. Explain the Importance of Text Editors in Linux.
Text editors like vi and nano are used to create and edit files, configuration scripts, and programs in Linux.

11. Explain Hard Link and Soft Link in Linux.
A hard link points directly to the inode of a file, while a soft link (symbolic link) points to the filename itself.

12. Restate the vi Command for Deleting the Current Word.
In the vi editor, the command dw deletes the word from the cursor position.

Part B

Short Essay Type Questions

Answer any 6 questions.

Each question carries 5 marks.

13. Describe the Lifecycle of a Thread.
A thread passes through states: New → Ready → Running → Waiting → Terminated. Each state transition depends on scheduling and synchronization events.

Definition-**1 mark**
Diagram-**2marks**
Explanation of each state **-2 marks**

14. Explain how the Bootstrapping Process works.
Bootstrapping (booting) initializes system hardware and loads the operating system kernel into memory from secondary storage.
Definition-**1mark**
Steps in boot process(BIOS,Bootloader,kernel load)**-4 marks**

15. What is meant by Interprocess Communication (IPC)?
IPC allows processes to exchange data or signals. Mechanisms include message queues, shared memory, and pipes.
Definition-**1 mark**
Methods-**3 marks**
Example or usecase**- 1 mark**

16. State the Difference between Safe and Unsafe State in Deadlock.
In a safe state, the system can allocate resources without causing a deadlock; in an unsafe state, deadlock may occur.
Definition-**2 marks**
Example-**2 marks**
Condition-**1 mark**

17. Explain how Paging works.
Paging divides both physical and logical memory into fixed-size blocks (frames and pages). The page table maps logical pages to physical frames.
Definition - **1 mark**
Diagram(logical to physical mapping)-**2 marks**
Explanation of steps-**2 marks**

18. Explain various Strategies for Handling Deadlock.
Deadlocks can be handled by prevention (avoid circular wait), avoidance (Banker's algorithm), detection, and recovery techniques.
Definition of deadlock- **1mark**
Deadlock prevention- **1 mark**
Deadlock Avoidance- **1 mark**
Detection and recovery-**2 marks**

19. Analyse the Role of the Kernel I/O Subsystem.
The kernel I/O subsystem manages communication between hardware devices and user processes through device drivers and buffering.
Definition of kernel I/O -**1 mark**
Role of device management- **2 marks**
Coordination with user space-**2 marks**

20. Explain the Hardware Requirements for Linux OS.
Linux requires a processor (x86 or ARM), RAM (minimum 512 MB recommended 2 GB+), storage, and basic peripherals to operate efficiently.
Processor- **1mark**
Memory requirements,storage requirements-**2 marks**
Peripheral and network support-**2 marks**

Part C

Essay Type Questions

Answer any 2 questions.

Each question carries 10 marks.

21. Write a brief description of First Come First Serve (FCFS) CPU Scheduling Algorithm.
FCFS is the simplest scheduling method where processes are executed in the order they arrive in the ready queue. It is non-preemptive and can lead to the “convoy effect.” Example: If processes arrive in order of burst time 10, 5, 2, the average waiting time can be high due to long jobs at the front.

Definition -**2 marks**
Explanation of FCFS Algorithm -**5 marks**
Advantages and Limitations-**3marks**
Total=**10 marks**

22. Apply the Concept of Virtual Memory to explain how a 32-bit Logical Address Space can be mapped onto a Smaller Physical Memory using Paging.
Virtual memory allows processes to use more memory than physically available by storing inactive pages on disk. A 32-bit address can reference 4 GB of virtual space, which is divided into pages mapped dynamically to physical frames using a page table.
Concept and Definition of virtual memory- **2marks**
Explanation of paging mechanism-**3 marks**
Diagram/Address-mapping example -**3 marks**
Benefits and practical uses- **2marks**
Total=**10 marks**

23. Differentiate between Programmed I/O, Interrupt-driven I/O, and DMA, and evaluate which is most efficient in High-Speed Data Transfer.

Programmed I/O: CPU actively waits for I/O completion.

Interrupt-driven I/O: Device sends an interrupt when ready, freeing CPU time.

DMA (Direct Memory Access): Data is transferred directly between memory and device without CPU involvement.
Most efficient: DMA, especially for large data transfers.
Definition of each I/O method-**3 marks**
Comparative table or differences-**3 marks**
Evaluation of efficiency with reason-**3 marks**
Suitable application example-**1 mark**
Total=**10 marks**

END OF SCHEME